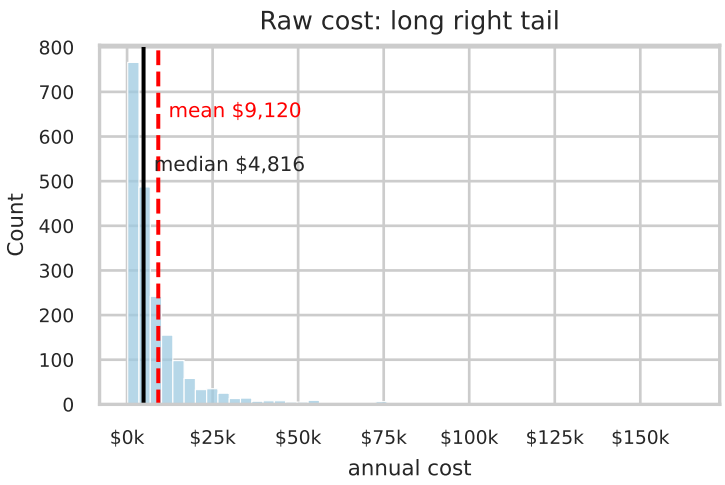
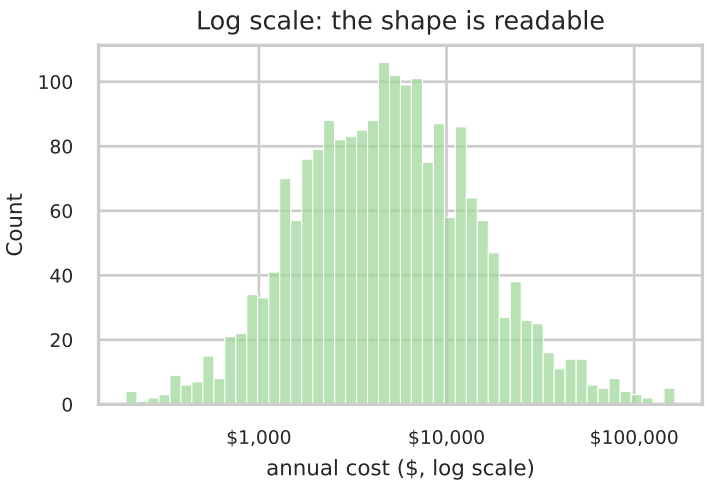
Day 5 Cheat Sheet: EDA with matplotlib & seaborn

Every question has a chart that answers it best

```
EDA CHECKLIST (before any model)
  1. Shape and dtypes: df.shape, df.dtypes
  2. Missing values: df.isna().mean()
  3. Duplicates: df.duplicated().sum()
  4. One variable at a time: histograms, counts
  5. Two at a time: box plots, scatter, heatmap
  6. Write 3-5 findings in plain words
```

QUESTION	CHART	SEABORN
How is one number distributed?	histogram	sns.histplot(x=..., log_scale=True)
How do groups compare?	box / violin	sns.boxplot(x="group", y="value")
How many in each category?	bar (counts)	sns.countplot(x="category")
How do two numbers relate?	scatter	sns.scatterplot(x=, y=, hue="third")
Which numbers move together?	heatmap	sns.heatmap(df.corr(), annot=True)
What changes over time?	line	sns.lineplot(x="month", y="count")
Many pairs at once (quick scan)?	pair plot	sns.pairplot(df, hue="smoker")





THIS DATA (simulated): 2,012 rows, 12 duplicates, so 2,000 patients | BMI missing 7.1%
Mean cost $9,120 vs. median $4,816 (right-skewed: report the median, plot on a log scale)
Median cost: smokers $11,117 vs. non-smokers $4,124 (about 2.7x)
Spearman with log cost: age 0.48, chronic conditions 0.45, BMI 0.13

CHART HABITS THAT GET YOU HIRED
- Title = the finding ("Smokers cost 2.7x more"), not the variable names
- Log scale for costs and lengths of stay; say so on the axis label
- Compare groups on the SAME axis; avoid pie charts and 3-D bars
- A gap between groups is a lead, not a cause: check for a third variable (age, plan)

```
import seaborn as sns, matplotlib.pyplot as plt
sns.set_theme(style="whitegrid")
sns.histplot(df["annual_cost"], bins=50, log_scale=True)
sns.boxplot(data=df, x="smoker", y="annual_cost", log_scale=True)
sns.scatterplot(data=df, x="age", y="annual_cost", hue="smoker", alpha=0.5)
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="RdBu_r", vmin=-1, vmax=1)
```

Infographic: One Patient Dataset, Six Questions

Simulated: 2,000 patients, age, plan, BMI, smoking, chronic conditions, ER visits, annual cost

Load	Inspect	Clean	One var	Two vars	Findings
read_csv	shape, dtypes	missing, duplicates	histograms, counts	box, scatter, heatmap	3-5 plain sentences

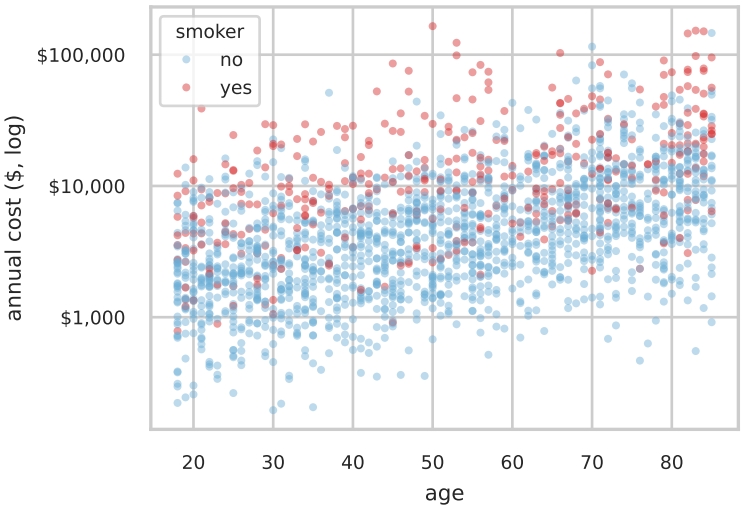
ROWS
2,012 to 2,000

BMI MISSING
7.1%

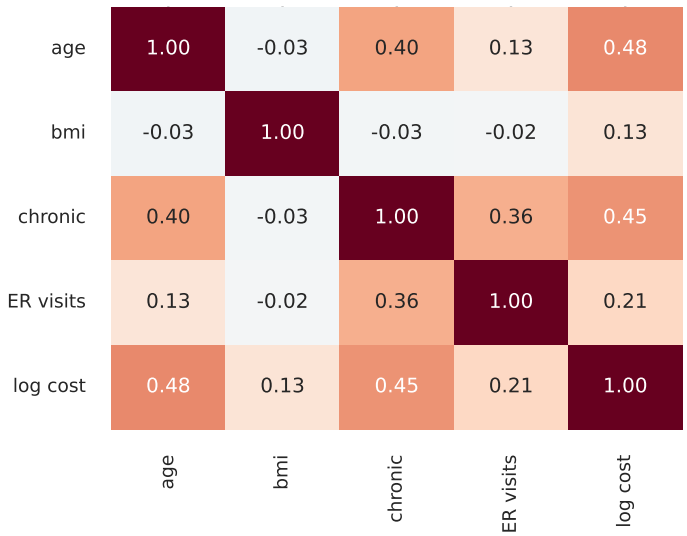
MEAN vs MEDIAN
$9.1k vs $4.8k

SMOKERS
2.7x median cost

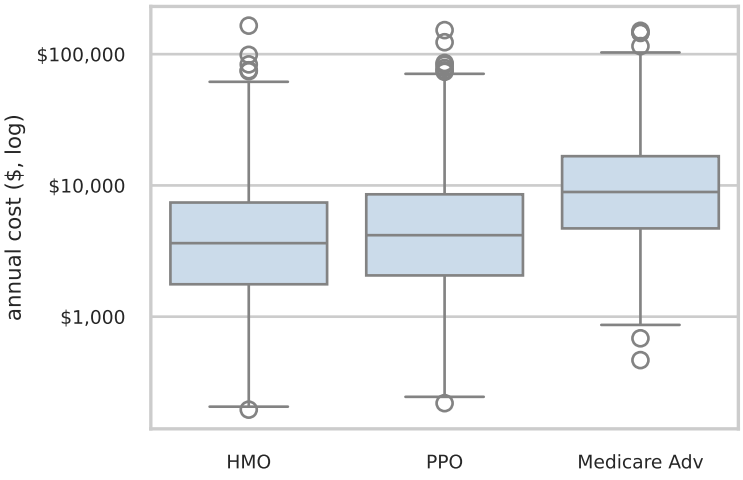
Two numbers: age vs. cost, colored by smoker



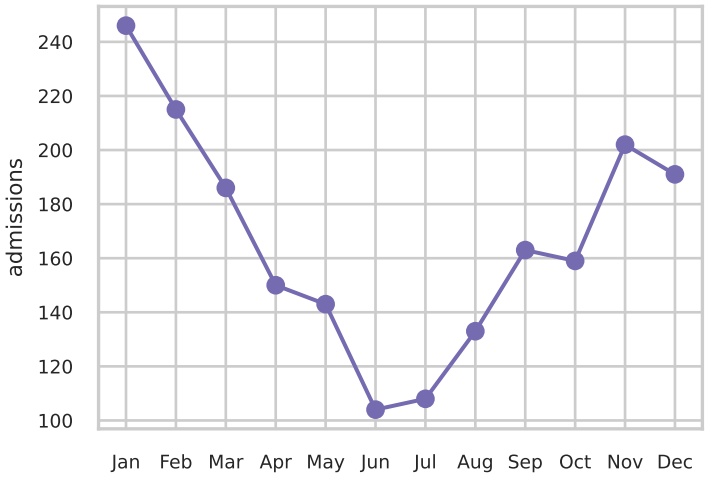
Move together: Spearman correlations



Groups: cost by plan (median age differs!)



Over time: admissions by month, 2025



Plan trap: Medicare Advantage looks most expensive (median $8,913 vs. HMO $3,632), but its members' median age is 75 vs. 43. Among patients 65+, PPO is actually higher. Always ask: what else differs?

Hands-On Worksheet: ds-genai-90-days / day05

60-90 minutes | notebook: day05/eda_patients.ipynb

[ ] mkdir day05, open VS Code, conda activate ds90, conda install seaborn

[ ] Build the simulated patient table (snippet 1) and save patients.csv

[ ] Inspect: shape, dtypes, share missing, duplicates (expect 2,012 rows, 12 dupes)

[ ] Drop duplicates and run describe(): compare mean vs. median annual cost

[ ] Histogram of annual cost, raw and log scale, side by side

[ ] Box plots: cost by smoker and cost by plan (log y). Note the medians

[ ] Scatter: age vs. cost, hue = smoker, log y

[ ] Spearman heatmap of age, BMI, chronic conditions, ER visits, log cost

[ ] Line chart: admissions by month. Which months peak?

[ ] Rewrite each chart title as a finding; add a 5-bullet findings cell

[ ] git add day05 && git commit -m 'Day 5: EDA with seaborn' && git push

[ ] Stretch: Snowflake: median cost by plan within age bands (see lesson)

```
import pandas as pd, numpy as np, seaborn as sns, matplotlib.pyplot as plt
df = pd.read_csv("patients.csv", parse_dates=["admit_month"])
print(df.shape, df.duplicated().sum()); df = df.drop_duplicates()
print(df.isna().mean().round(3))
print(df["annual_cost"].agg(["mean", "median"]))
sns.boxplot(data=df, x="smoker", y="annual_cost", log_scale=True); plt.show()
print(df.groupby("smoker")["annual_cost"].median())   # expect ~4,124 vs ~11,117
young_old = pd.cut(df["age"], [17, 44, 64, 85])
print(df.groupby([young_old, "plan"], observed=True)["annual_cost"].median().unstack())
```

Rows after dedup: _____ BMI missing: _____% Mean cost: $_____ Median: $_____

Smoker median: $_____ Non-smoker: $_____ Strongest Spearman with log cost: _____

Day 5 Self-Check Quiz

1. Name the first three checks you run on a new dataset, before any chart.
_____

2. Which chart answers 'How do readmission rates compare across 5 hospitals?'
_____

3. Mean cost is $9,120 and median is $4,816. What does that tell you about the shape?
_____

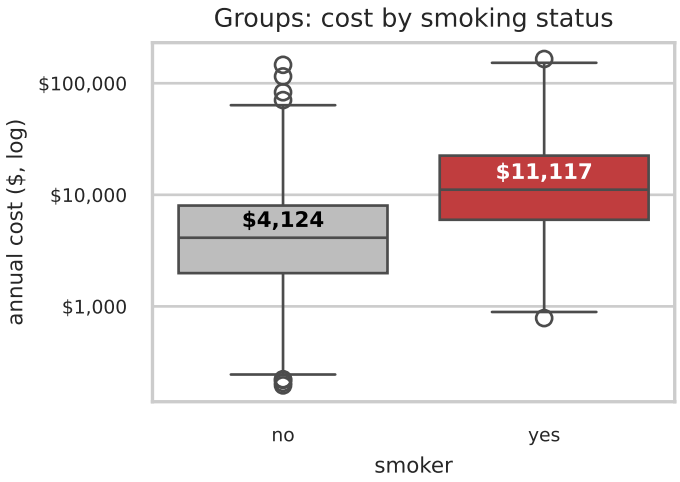
4. Why plot annual cost on a log scale? What must you say on the axis?
_____

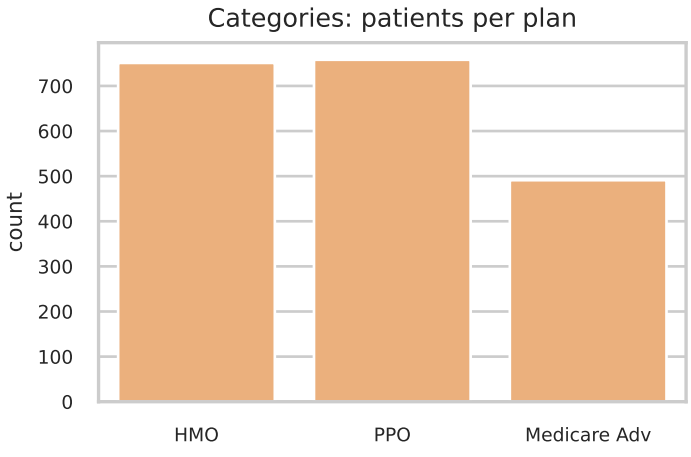
5. You want to see how age AND smoking relate to cost in one chart. Which chart?
_____

6. Medicare Advantage has the highest median cost. Name one reason not to blame the plan.
_____

7. When would you use Spearman instead of Pearson correlation?
_____

8. Rewrite the title 'annual_cost by smoker' as a finding.
_____





Answers:
1 shape/dtypes, missing values, duplicates 2 bar chart of rates (box plot if you have patient-level values)
3 right-skewed: a few very expensive patients pull the mean up 4 so the long tail does not squash everyone else; label the axis 'log scale'
5 scatter of age vs. cost with hue = smoker 6 its members are much older (median age 75 vs. 43), so age confounds plan
7 skewed data, outliers, or a curved but always-increasing relationship (Spearman uses ranks)
8 e.g. "Smokers' median annual cost is 2.7x non-smokers'" (simulated data)